

# Flash ADCs

Nan Sun  
Tsinghua University  
nansun@Tsinghua.edu.cn

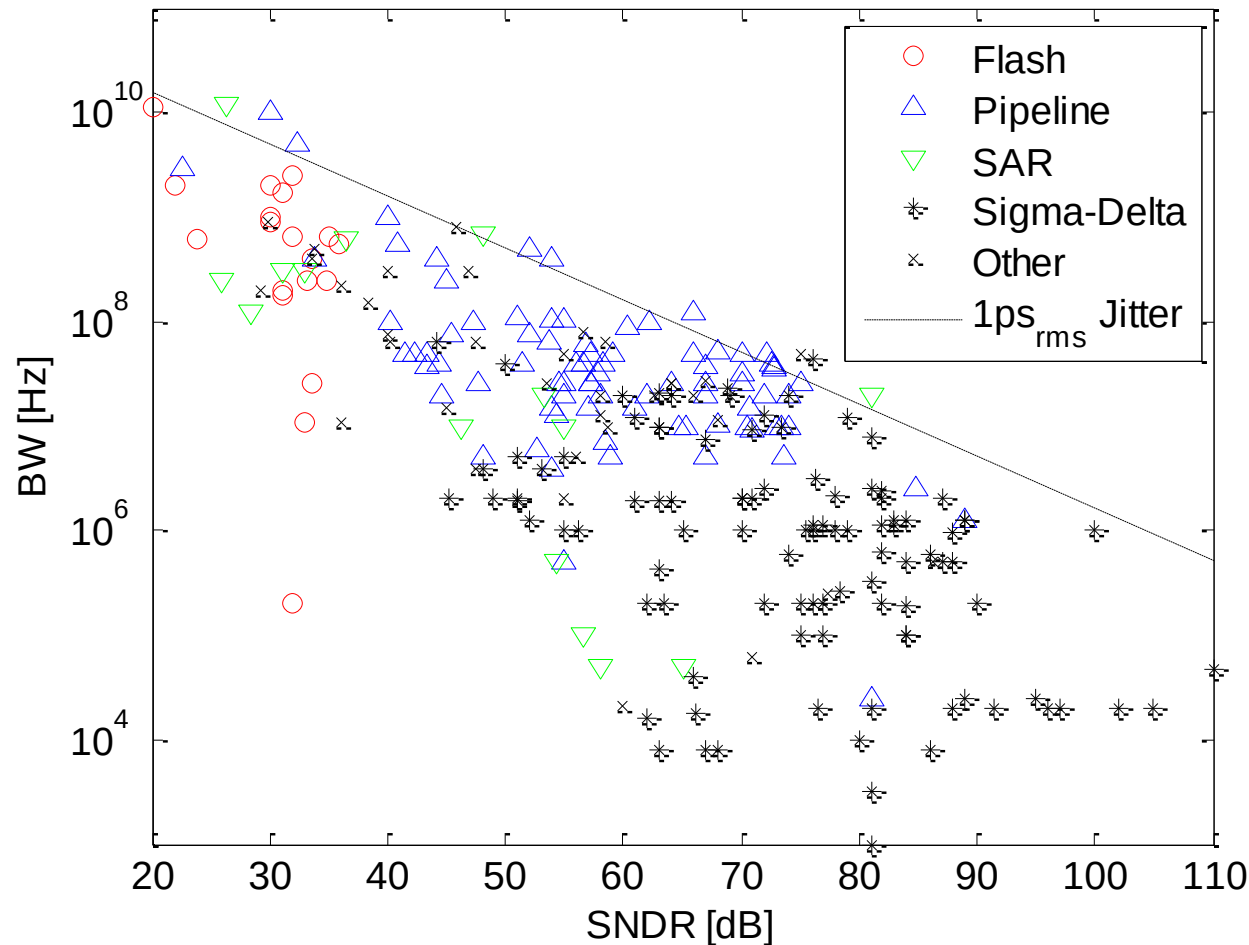
# Nyquist ADC Architectures

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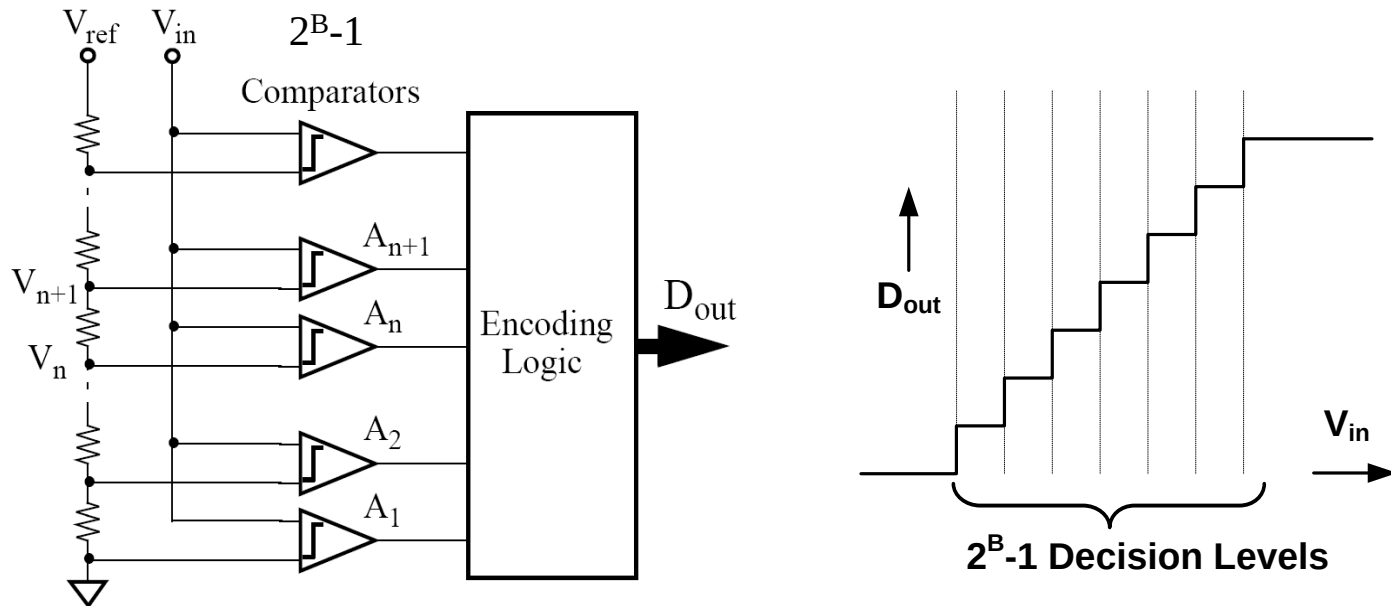
- Nyquist rate
  - Flash ADC
    - Instantaneous comparison with  $2^B-1$  reference levels
  - Folding and interpolating ADC
    - Two steps
  - Pipeline ADCs
    - Multi steps
  - Successive approximation ADCs
    - Conversion via a binary search algorithm
  - Single or dual slope ADCs
    - Input is converted by measuring the time it takes to charge/discharge a capacitor from/to input voltage

# ADC Performance Survey (ISSCC & VLSI 97-08)

Data: <http://www.stanford.edu/~murmman/adcsurvey.html>

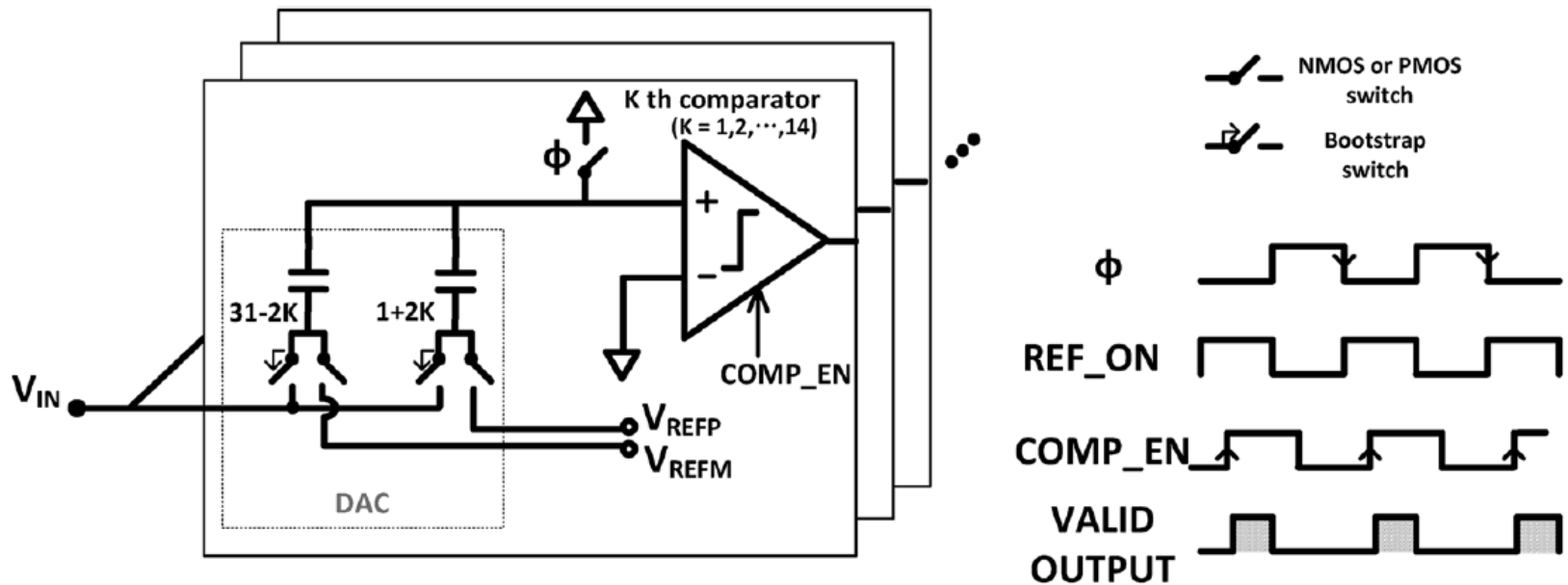


# Flash ADC



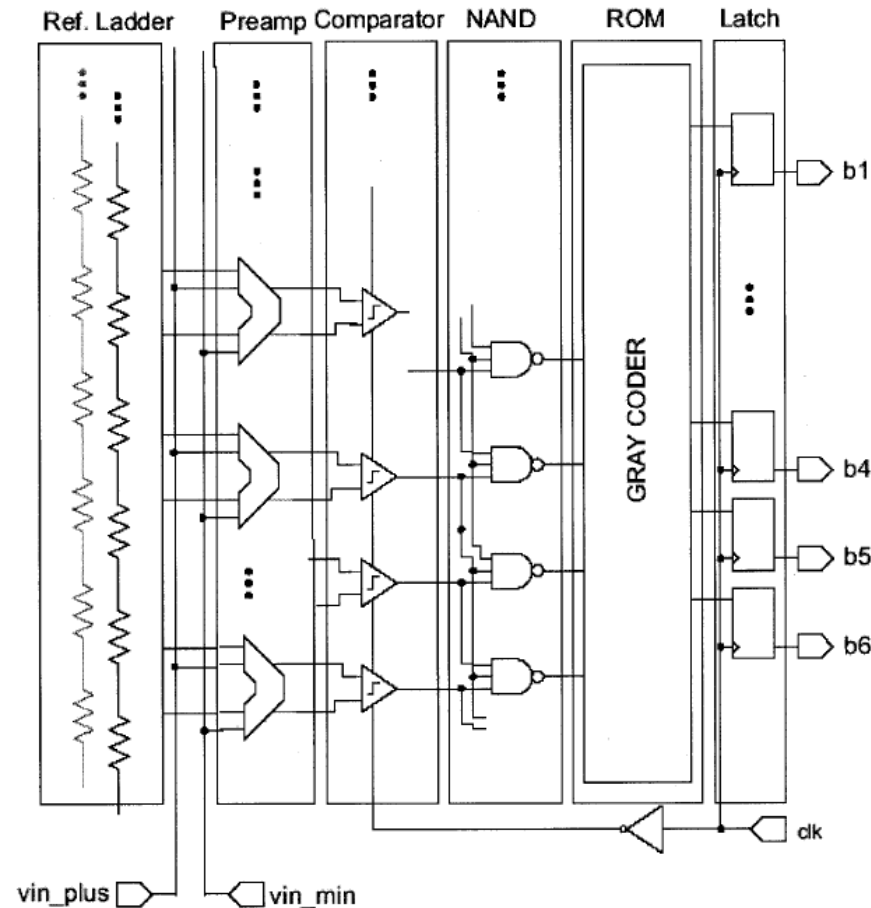
- Fast speed 😊
  - Speed limited by 1 comparator plus encoding logic
- High complexity ( $2^B - 1$  comparators) and large input capacitance 😞
  - Typically use for resolution up to 6 bits

# Use Capacitor to Generate References



S. Lee, A. Chandrakasan, and H.-S. Lee, JSSC, Dec. 2014.

# Real Implementation



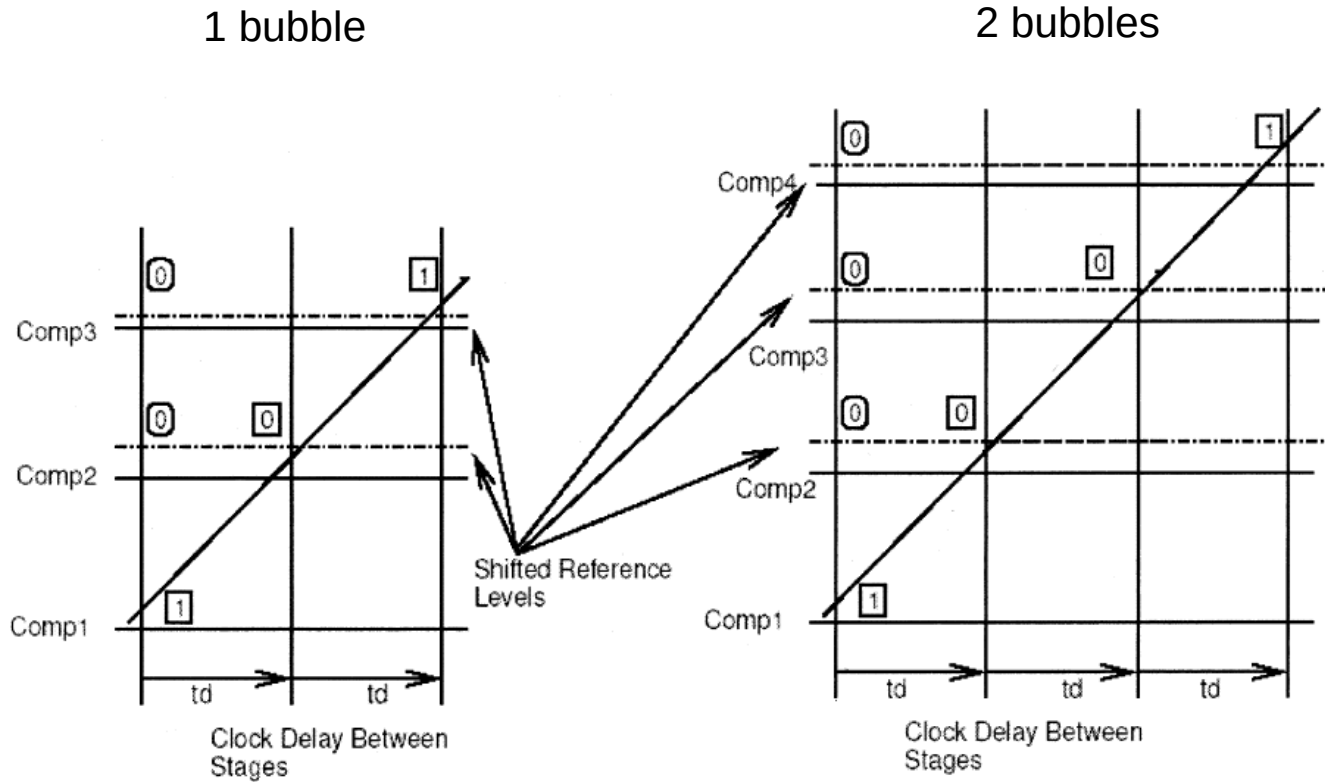
K. Uyttenhove, M.S.J. Steyaert, "A 1.8-V 6-bit 1.3-GHz flash ADC in 0.25- $\mu$ m CMOS," *Solid-State Circuits, IEEE JSSC*, pp. 1115-1122, July 2003.

# Error Sources

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- Comparator
  - Offset
  - Nonlinear input capacitance
  - Kickback noise (disturb reference)
  - Metastability
- Timing skew
  - Comparators clocked at different times
  - Produce bubbles that lead to sparkle codes

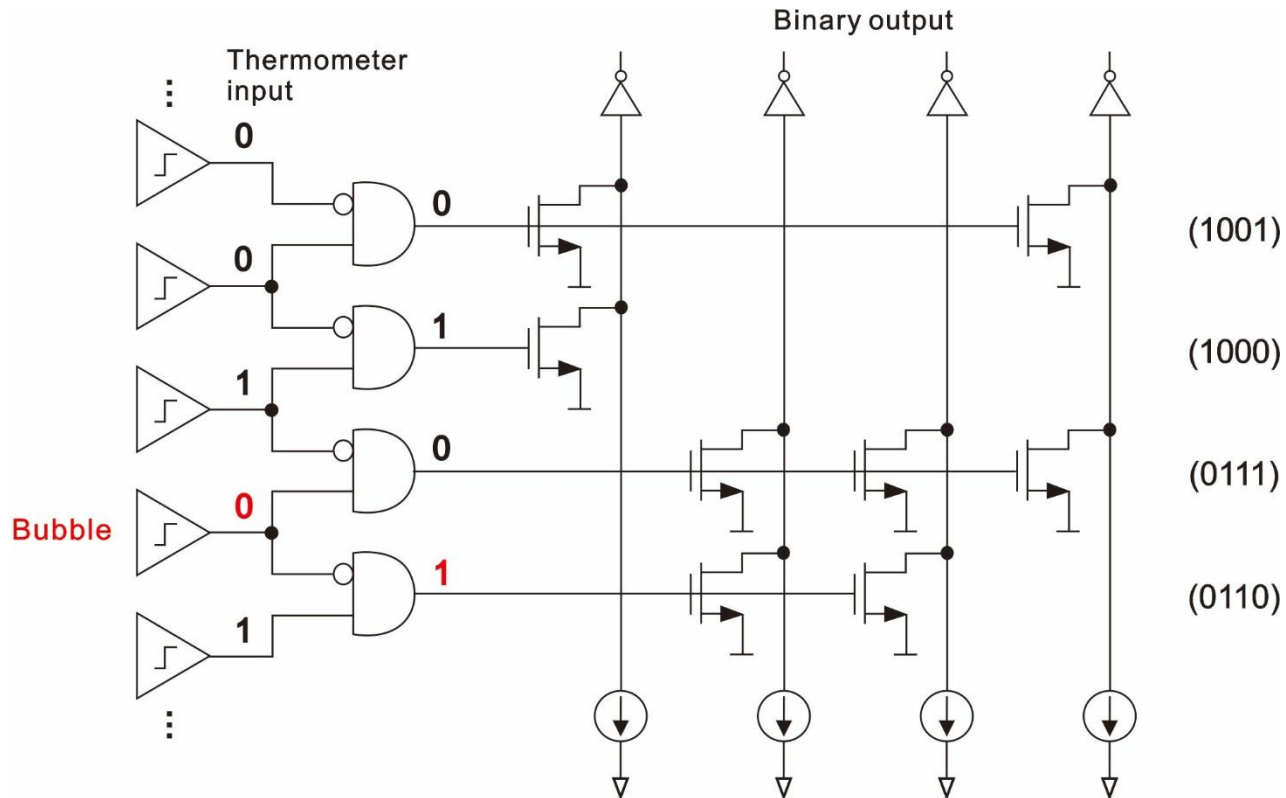
# Bubbles Due to Timing Errors



K. Uyttenhove, M.S.J. Steyaert, "A 1.8-V 6-bit 1.3-GHz flash ADC in 0.25- $\mu$ m CMOS," *Solid-State Circuits, IEEE JSSC*, pp. 1115-1122, July 2003.

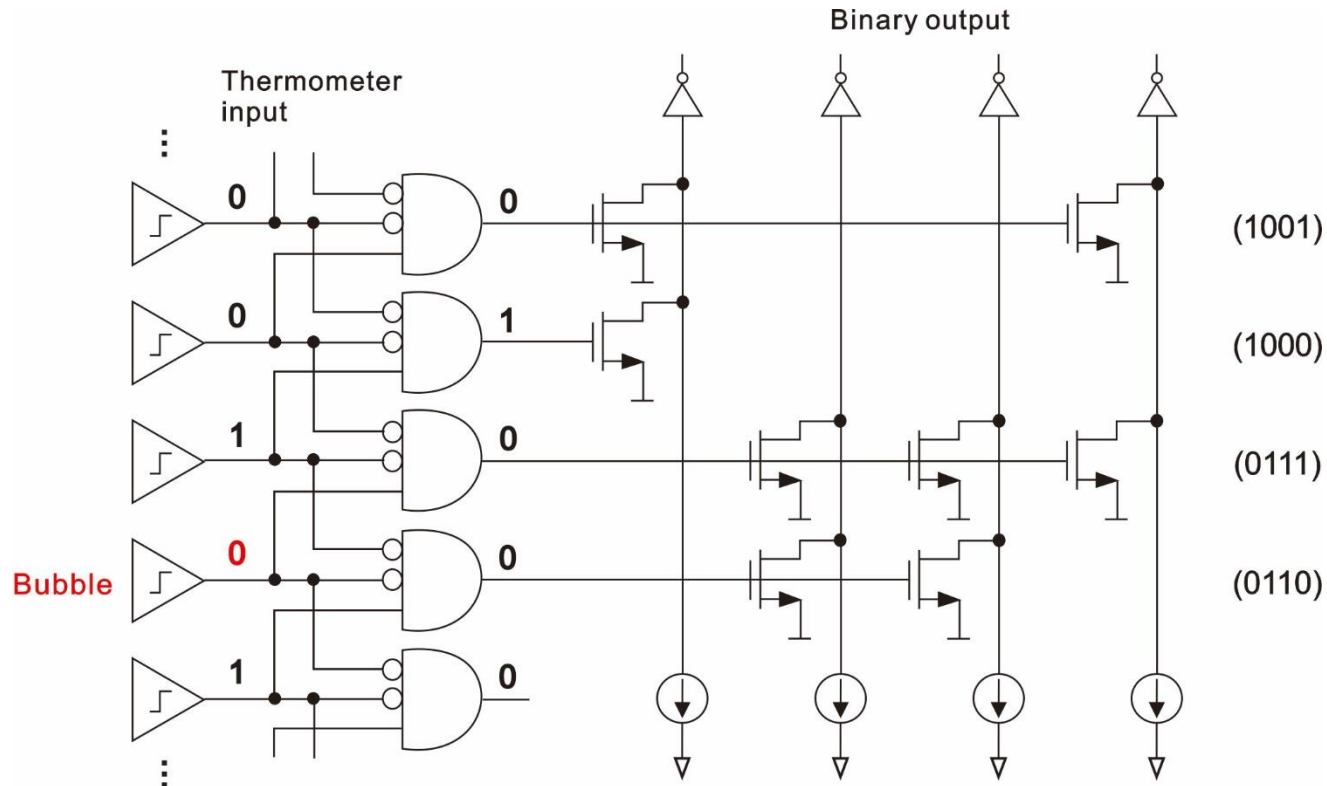


# Bubble Problem



- Two rows in the ROM are enabled.
- Correct output: 1000, actual output: 1110

# Bubble Tolerant Encoder



- Protects against isolated bubbles
- Reference: C. W. Mangelsdorf, "A 400-MHz Flash Converter with Error Correction," IEEE JSSC., pp. 184-191, Feb. 1990.

# Wallace Encoder (“Ones Adder”)

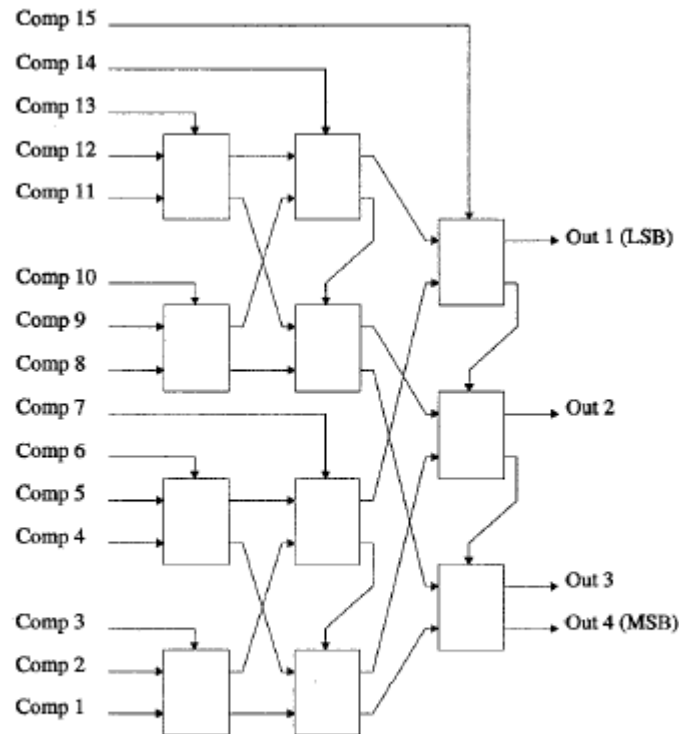


Fig. 1. Example of a Wallace tree encoder for a 4-bits flash converter

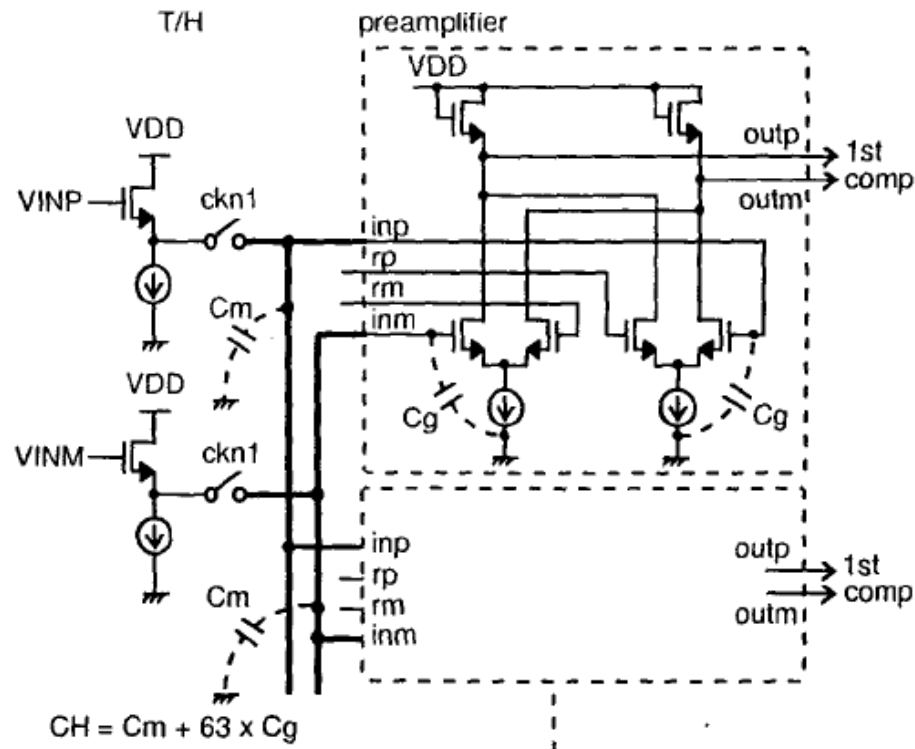
F. Kaess, R. Kanan, B. Hochet, and M. Declercq, “New encoding scheme for high-speed flash ADC’s,” IEEE ISCAS, pp. 5–8, 1997.

Number of adders required for N bits

$$\sum_{i=1}^N (i-1)2^{N-i}$$

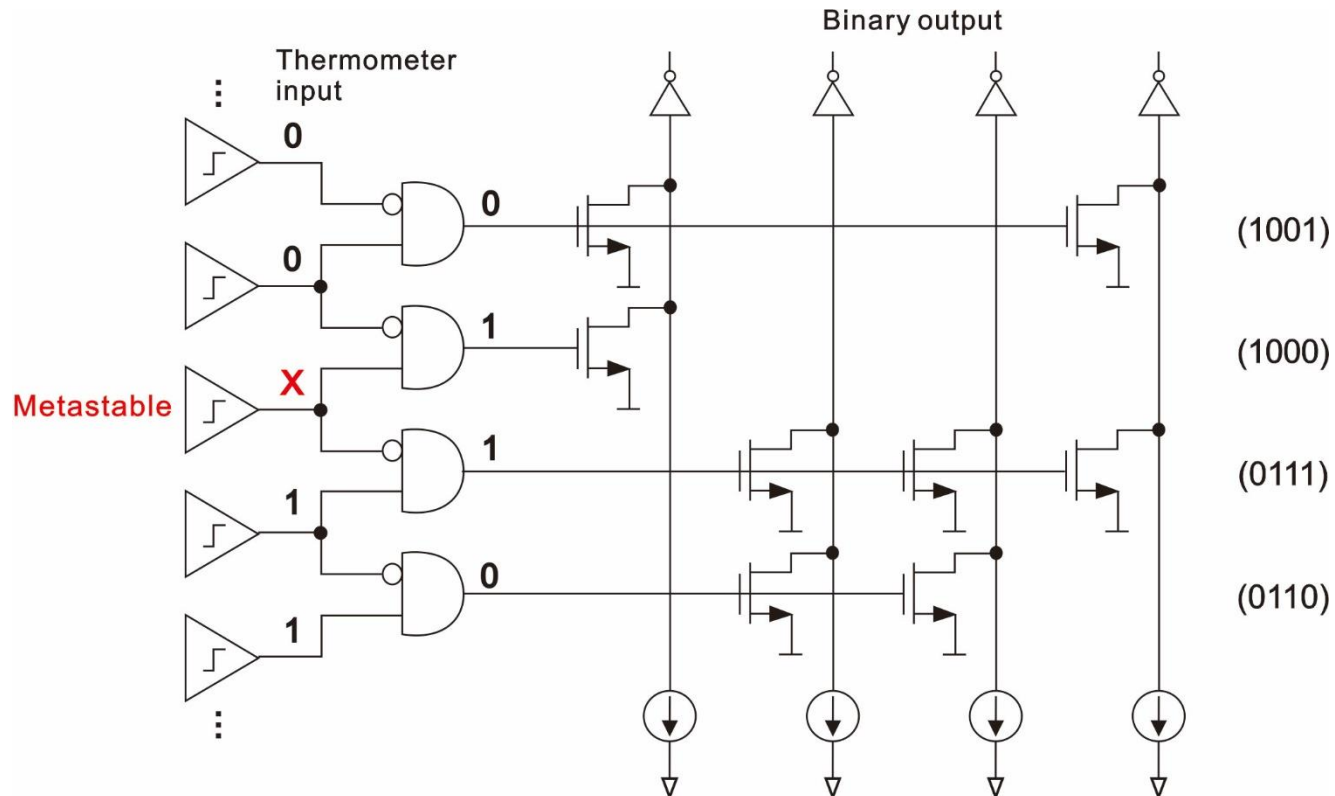
# Add Track-and-Hold

- Completely remove the timing skew problem ☺
- Speed and power penalty ☹



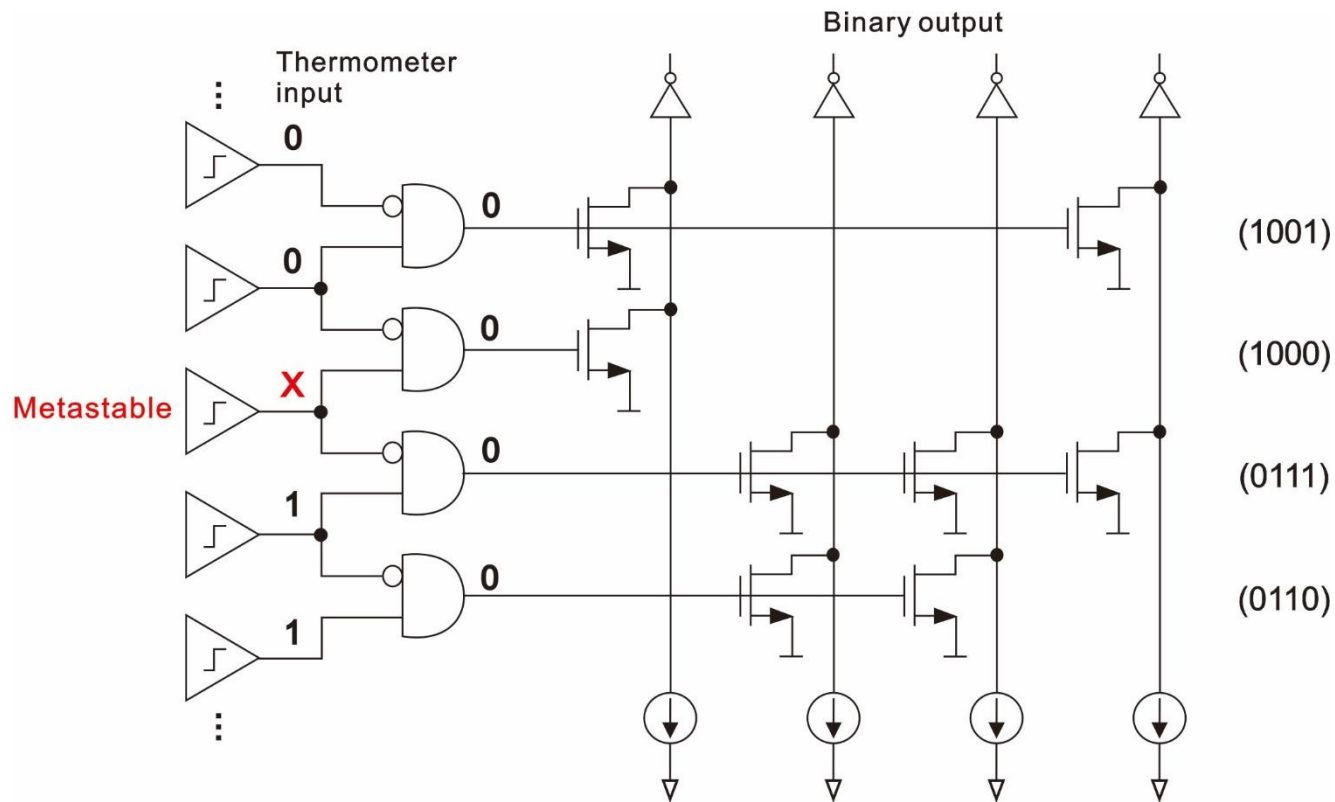
Y. Tamba, and K. Yamakido, "A CMOS 6 b 500 MSample/s ADC for a hard disk drive read channel," *ISSCC Dig. Techn. Papers.* pp.324-325, Feb. 1999.

# Metastability Problem



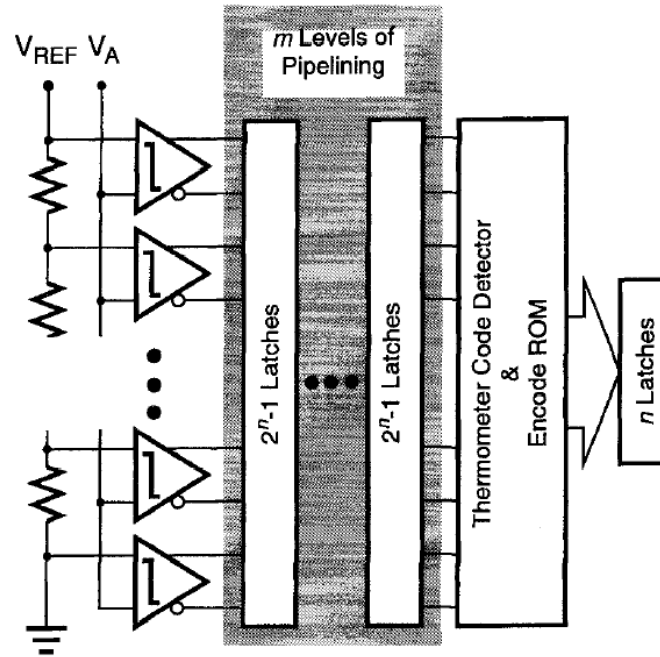
- Metastability happens when X drives multiple gates
- Different gates interpret X differently due to mismatches
- Correct output: 0111 or 1000, actual output: 1111

# Another Possibility



- Correct output: 0111 or 1000, actual output: 0000

# Solution 1: Latch Pipelining

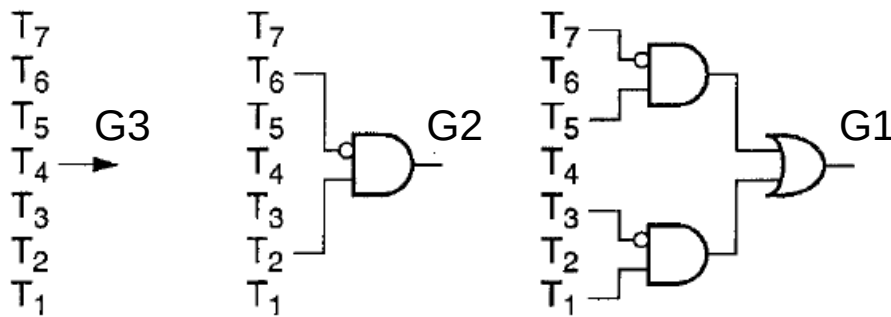


- Use additional latches to create extra gain before the encoder
- Latency, not delay
- Power hungry and area inefficient

# Solution 2: Gray Encoding

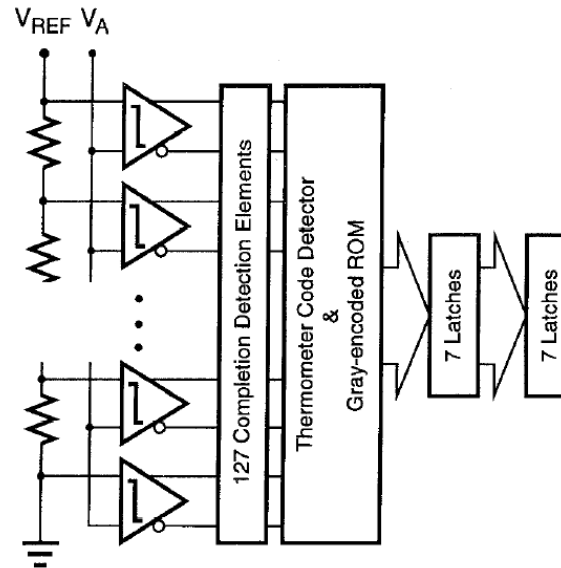
| Thermometer Code |                |                |                |                |                |                | Gray           |                |                | Binary         |                |                |
|------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| T <sub>1</sub>   | T <sub>2</sub> | T <sub>3</sub> | T <sub>4</sub> | T <sub>5</sub> | T <sub>6</sub> | T <sub>7</sub> | G <sub>3</sub> | G <sub>2</sub> | G <sub>1</sub> | B <sub>3</sub> | B <sub>2</sub> | B <sub>1</sub> |
| 0                | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |
| 1                | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 0              | 0              | 1              |
| 1                | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 1              | 0              | 1              | 0              |
| 1                | 1              | 1              | 0              | 0              | 0              | 0              | 0              | 1              | 0              | 0              | 1              | 1              |
| 1                | 1              | 1              | 1              | 0              | 0              | 0              | 1              | 1              | 0              | 1              | 0              | 0              |
| 1                | 1              | 1              | 1              | 1              | 0              | 0              | 1              | 1              | 1              | 1              | 0              | 1              |
| 1                | 1              | 1              | 1              | 1              | 1              | 0              | 1              | 0              | 1              | 1              | 1              | 0              |
| 1                | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 0              | 0              | 1              | 1              | 1              |

- Each T<sub>i</sub> affects only one G<sub>i</sub>
  - Avoids inconsistent interpretation
- Also helps protect against sparkles





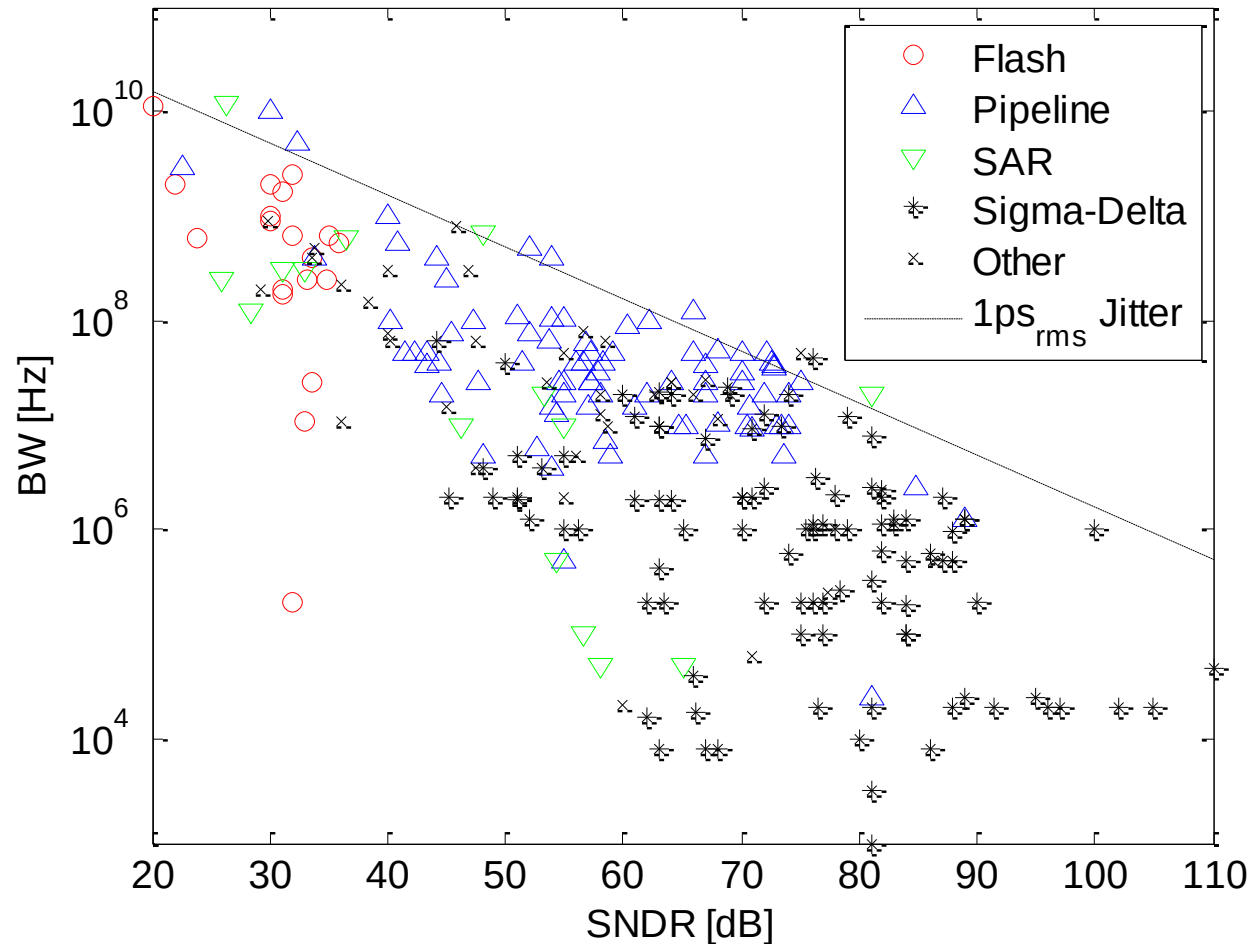
# Efficient Implementation



- Reference
  - C. Portmann and T. Meng, “Power-efficient metastability error reduction in CMOS flash A/D converters,” IEEE JSSC, pp. 1132-40 , Aug. 1996.

# Why Low Resolution?

Data: <http://www.stanford.edu/~murmnn/adcsurvey.html>



# Comparator Offset

- Want comparator offset smaller than LSB/4
  - Budget half of the offset for the latch, the other half for the pre-amp → pre-amp offset must be smaller than  $\text{LSB}/4/\sqrt{2}$

$$3\sigma_{VOS} = 3 \frac{A_{VT}}{\sqrt{WL}} < \frac{1}{4\sqrt{2}} \frac{FSR}{2^B}$$

- E.g. 6-bit flash ADC,  $FSR = 1V$

$$3 \frac{A_{VT}}{\sqrt{WL}} < \frac{1}{4\sqrt{2}} \frac{1V}{2^6} = 2.8mV$$

$$WL > \left( \frac{3A_{VT}}{2.8mV} \right)^2 = \left( \frac{3 \cdot 4mV\mu m}{2.8mV} \right)^2 = 18.4\mu m^2 \Rightarrow W > \frac{18.4\mu m^2}{0.18\mu m} = 102\mu m$$

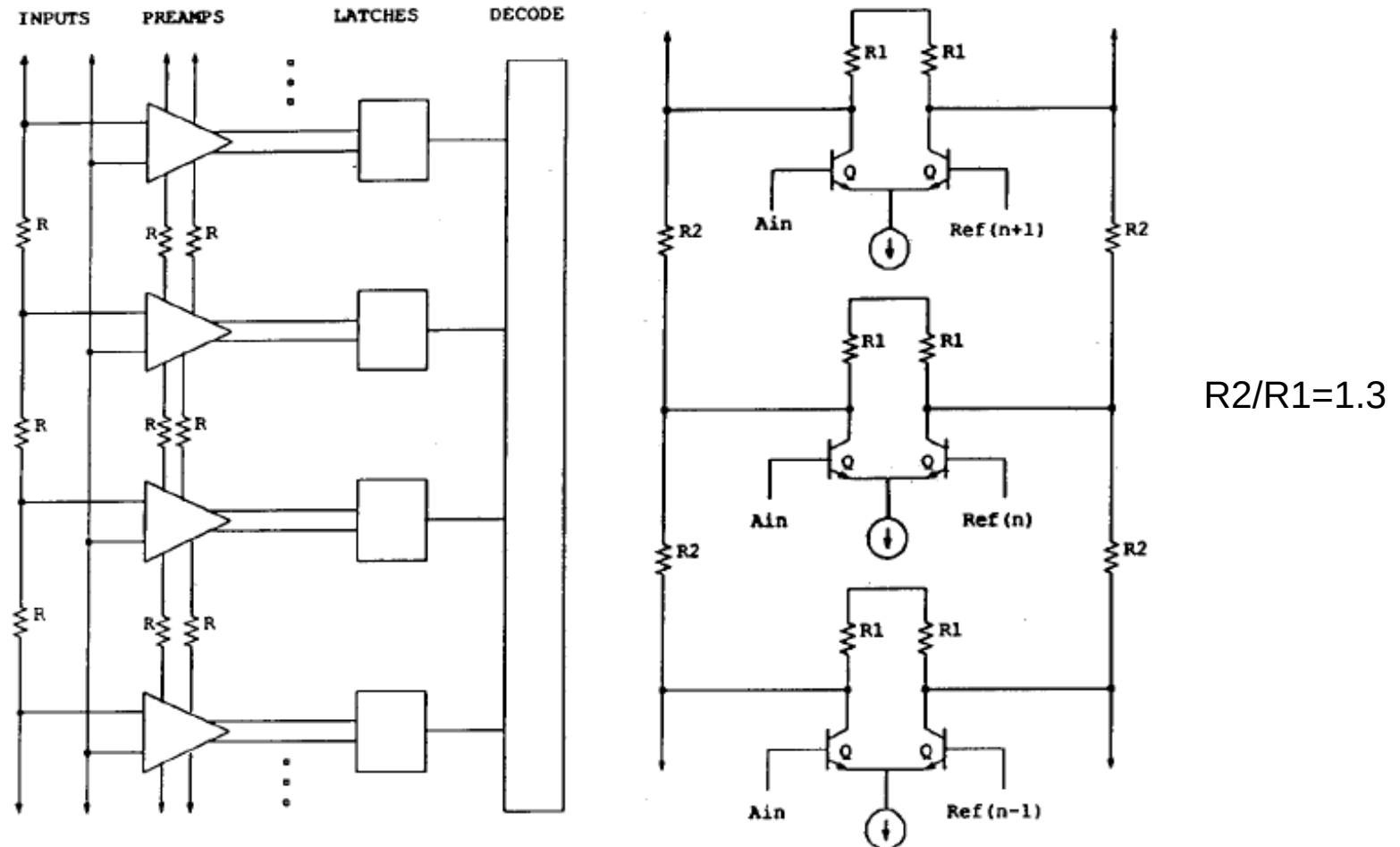
Does scaling  
help? Scaling  
reduced  $A_{VT}$

# Solutions

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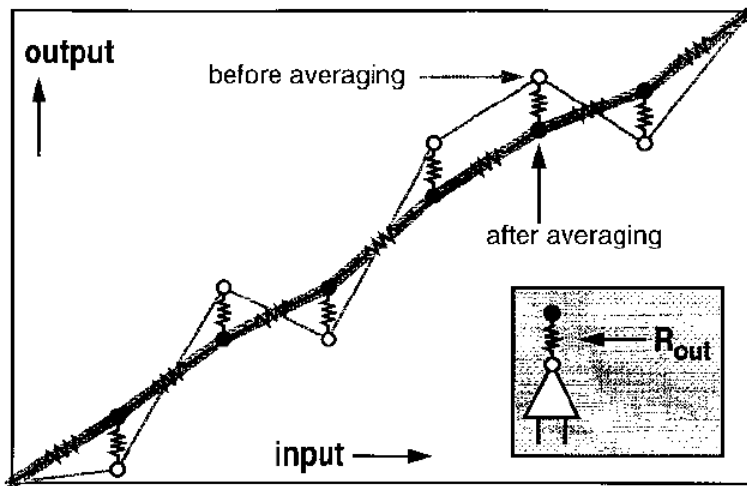
- Simply use large devices
  - For each extra bit
    - Reduce offset by 2x → increase device width by 4x
    - Increase comparator number by 2x
    - Overall 8x in power
- Input/output series offset cancellation
  - Reduce speed
- Offset averaging
- Calibration
- Comparator redundancy

# Offset Averaging (1)

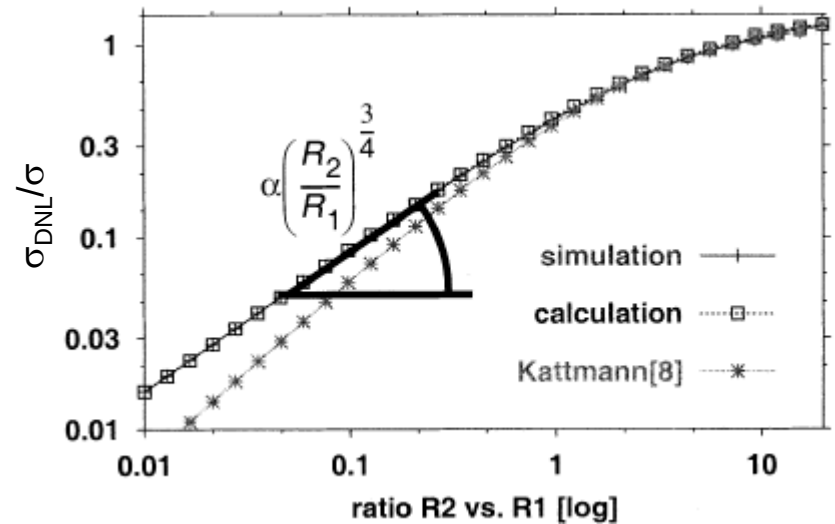
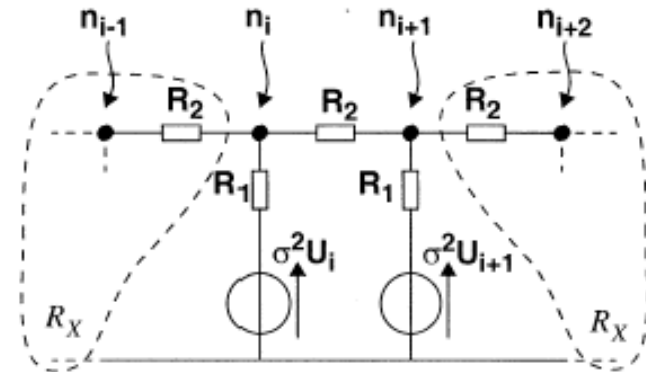


[Kattmann & Barrow, ISSCC 1991]

# Offset Averaging (2)



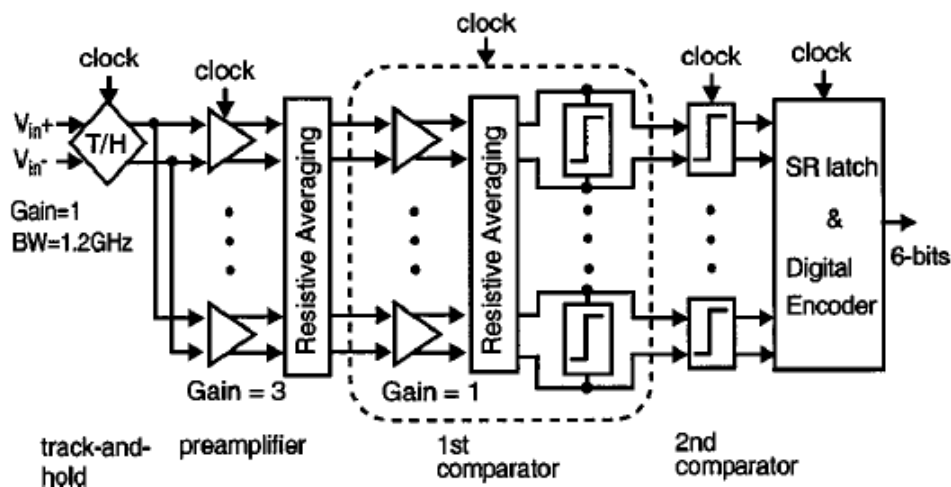
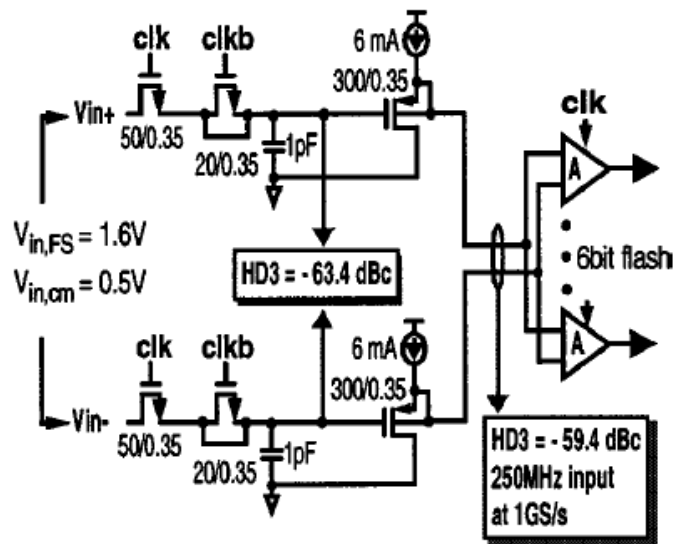
[Bult & Buchwald, JSSC 12/1997]



[Scholtens & Vertregt, JSSC 12/2002]

# 6-bit Flash ADC with Averaging

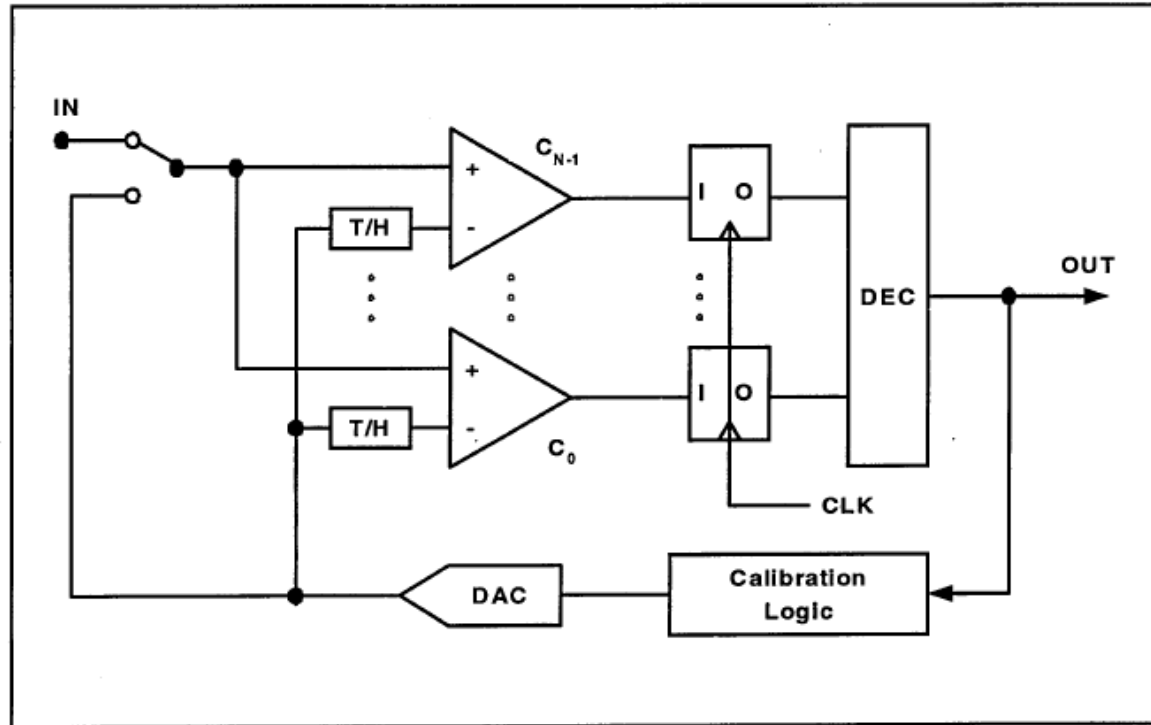
[Choi & Abidi, JSSC 12/2001]



|                                                   |                                                 |
|---------------------------------------------------|-------------------------------------------------|
| Resolution                                        | 6 bits                                          |
| Conversion Rate                                   | 1.3 GS/s                                        |
| INL/DNL @ $F_s = 1$ GHz                           | 0.35 LSB / 0.2 LSB                              |
| Input Range                                       | 1.6 $V_{p-p}$ differential                      |
| Input Capacitance (T/H)                           | 1 pF                                            |
| Bit Error Rate @ $f_s = 1$ GHz                    | $< 10^{-10}$                                    |
| SFDR @ $f_{in} = 100$ MHz                         | $> 44$ dB up to 1.3 GS/s                        |
| SNDR @ $f_{in} = 630$ MHz<br>@ $f_{in} = 650$ MHz | 35 dB @ 1 GS/s<br>32 dB @ 1.3 GS/s              |
| Power Dissipation<br>(50% due to logic and clock) | 500 mW @ 1 GS/s<br>545 mW @ 1.3 GS/s            |
| Voltage Supply                                    | 3.3 V                                           |
| Active/Total Die Area                             | 2x0.4 mm <sup>2</sup> / 2.2x2.2 mm <sup>2</sup> |
| Technology                                        | 0.35-μm CMOS                                    |

Averaging networks designed to reduce input referred offset by 3x

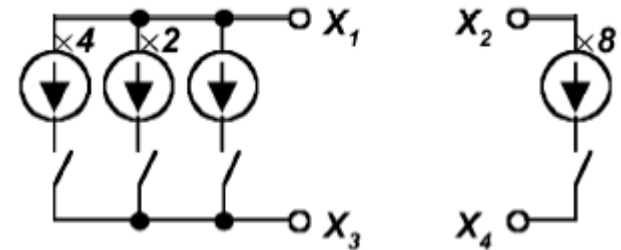
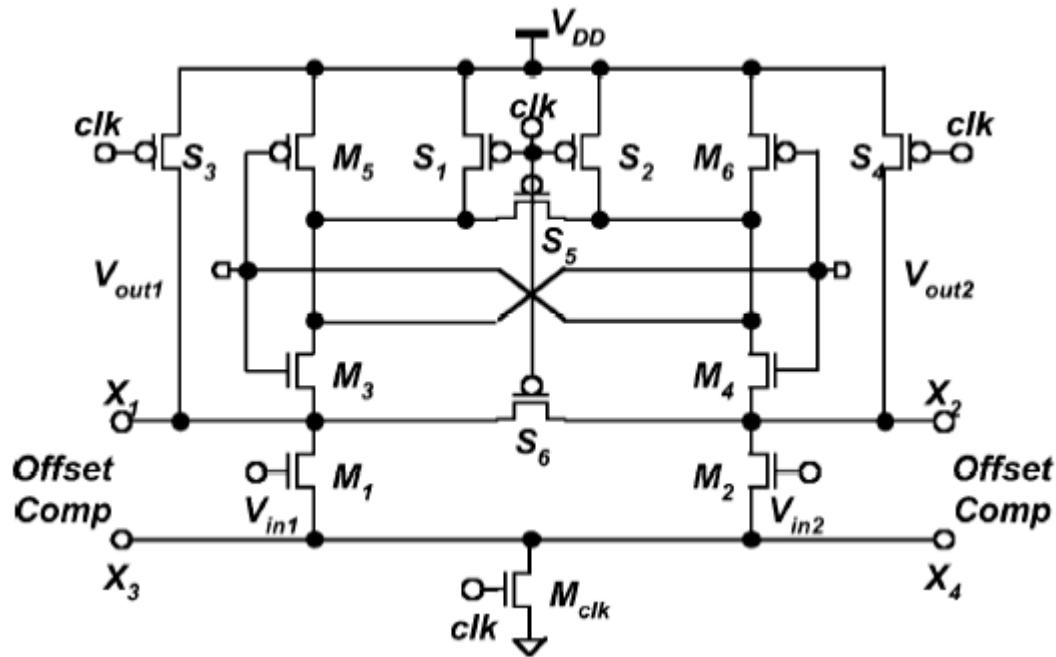
# Offset Calibration Example #1



S. Sutardja, "360 Mb/s (400 MHz) 1.1 W 0.35 $\mu$ m CMOS PRML read channels with 6 burst 8-20 $\times$  over-sampling digital servo," ISSCC Dig. Techn. Papers, Feb. 1999.

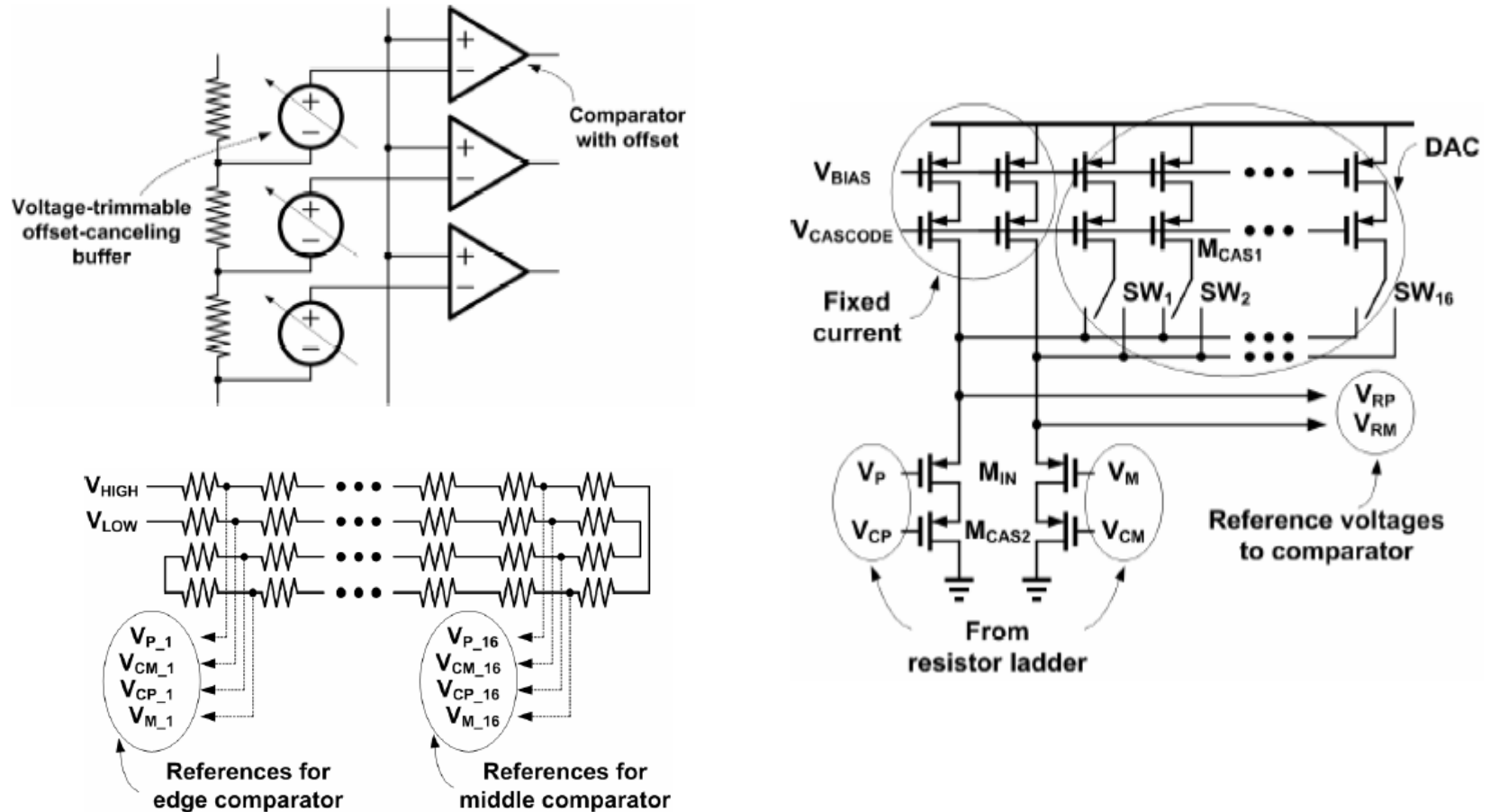


# Offset Calibration Example #2



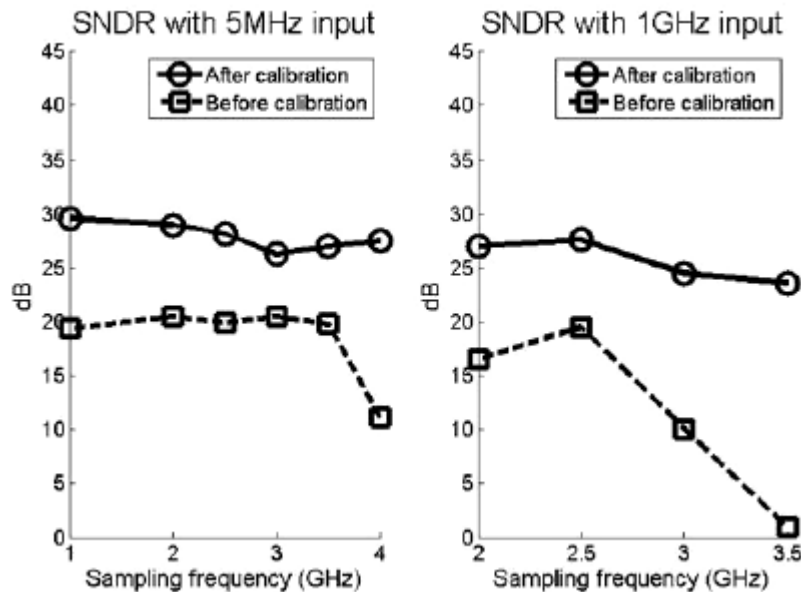
K.-L.J. Wong and C.-K.K. Yang, "Offset compensation in comparators with minimum input-referred supply noise," JSSC, May 2004.

# Offset Calibration Example #3 (1)



[Park & Flynn, "A 3.5 GS/s 5-b Flash ADC in 90 nm CMOS," CICC 2006]

# Offset Calibration Example #3 (2)

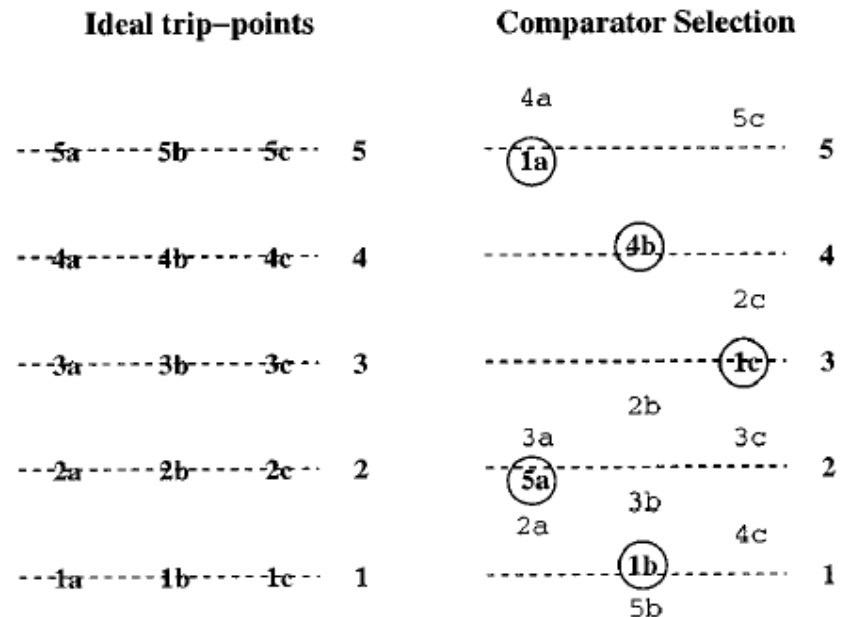
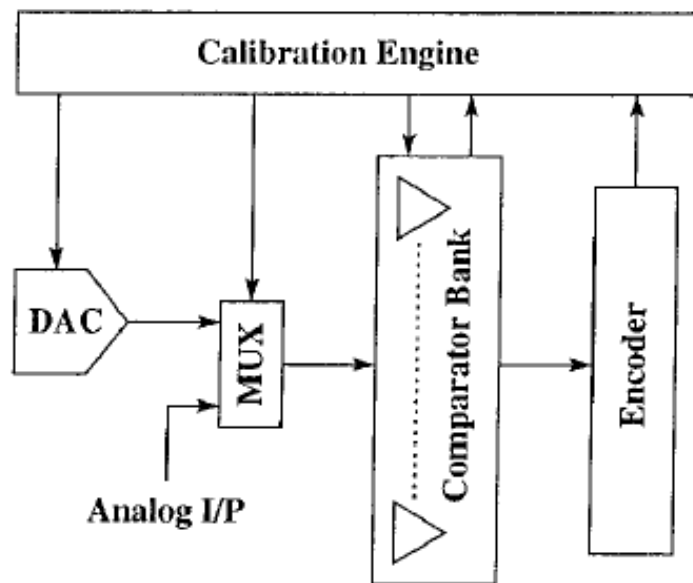


| ADC PERFORMANCE   |                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------|
| Technology        | CMOS 90 nm                                                                                                          |
| Resolution        | 5 bits                                                                                                              |
| Supply            | 1.4 V analog, 1.4 V digital, 1.8 V clock buffer                                                                     |
| Input range       | $\pm 320$ mV (LSB = 20 mV)                                                                                          |
| Sampling rate     | Up to 4 GS/s                                                                                                        |
| Power             | 227 mW<br>(115 mW: comparators, resistor ladder, bias.<br>17 mW: D-FFs, encoder, decimator.<br>95 mW: clock buffer) |
| DNL @ 3.5 GS/s    | -0.83 LSB ~ 0.93 LSB (after calibration)<br>-1.00 LSB ~ 4.51 LSB (before calibration)                               |
| INL @ 3.5 GS/s    | -0.89 LSB ~ 0.88 LSB (after calibration)<br>-2.20 LSB ~ 1.98 LSB (before calibration)                               |
| SNDR              | 27.5 dB @ 4.0 GS/s, 5 MHz input<br>23.6 dB @ 3.5 GS/s, 1 GHz input                                                  |
| Active area       | 0.658 mm <sup>2</sup> (including resistor ladder)                                                                   |
| Input capacitance | 540 fF                                                                                                              |
| Package           | Bare-die on board                                                                                                   |

[Park & Flynn, "A 3.5 GS/s 5-b Flash ADC in 90 nm CMOS," CICC 2006]

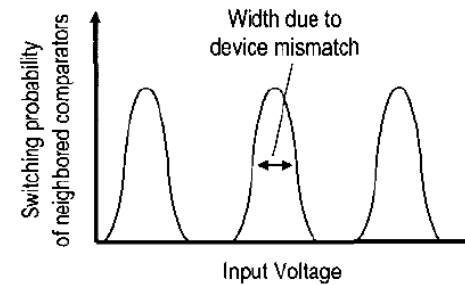
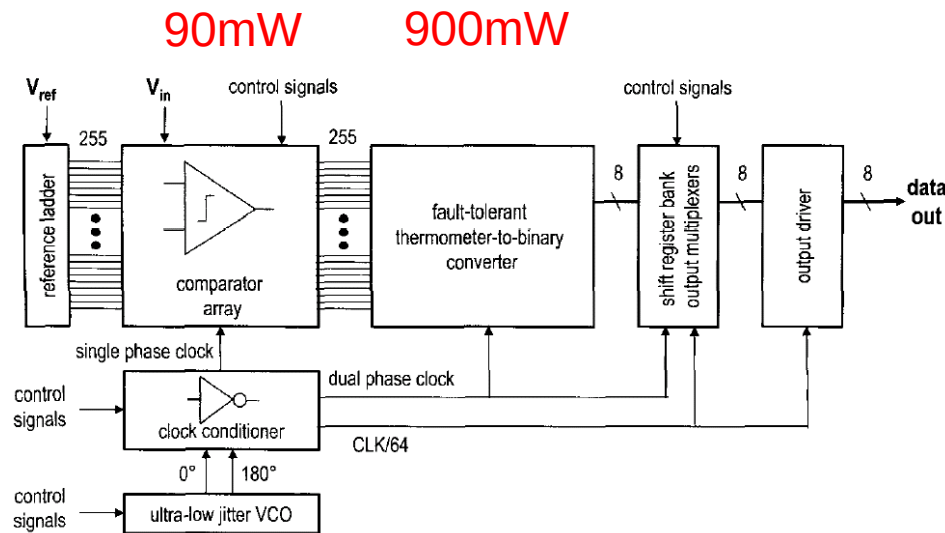
# Comparator Redundancy (1)

- Idea: build "a sea of imprecise comparators", then select which ones to use

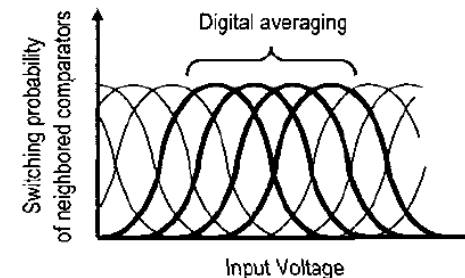


C. Donovan, M. P Flynn, "A 'digital' 6-bit ADC in 0.25- $\mu$ m CMOS," IEEE J. Solid-State Circuits, pp. 432-437, March 2002.

# Comparator Redundancy (2)



a)  
Classical  $n$  bit flash ADC:  
Use of  $2^n - 1$  precise comparators

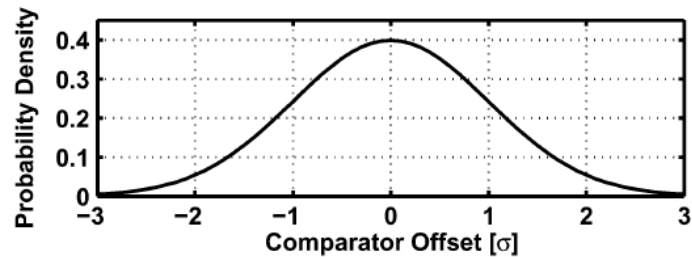


b)  
This  $n$  bit flash ADC:  
Use of  $4 \times 2^{n-1}$  comparators with overlapping decision levels

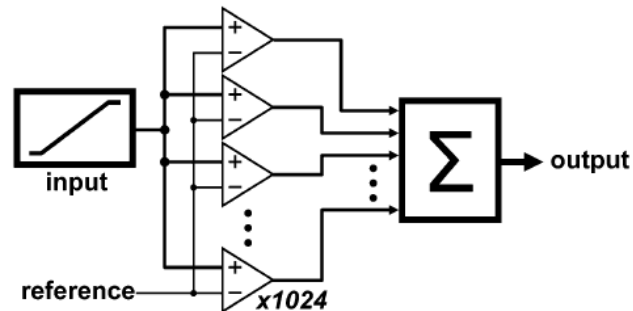
Fig. 1 Illustration of the ADC operation scheme.

Paulus et al., "A 4GS/s 6b flash ADC in 0.13 $\mu$ m CMOS," *VLSI Circuits Symposium*, 2004

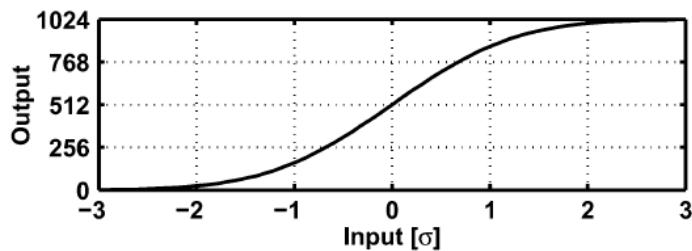
# Stochastic Flash ADC (1)



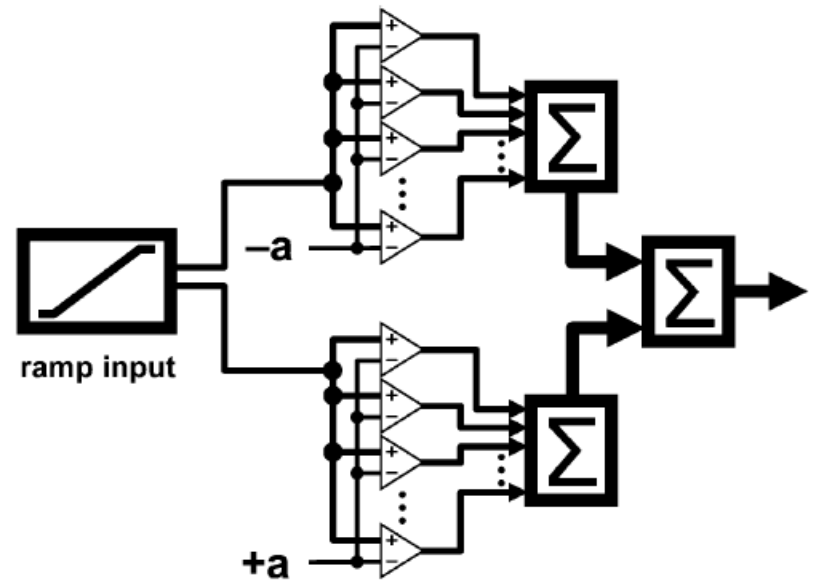
(a)



(b)

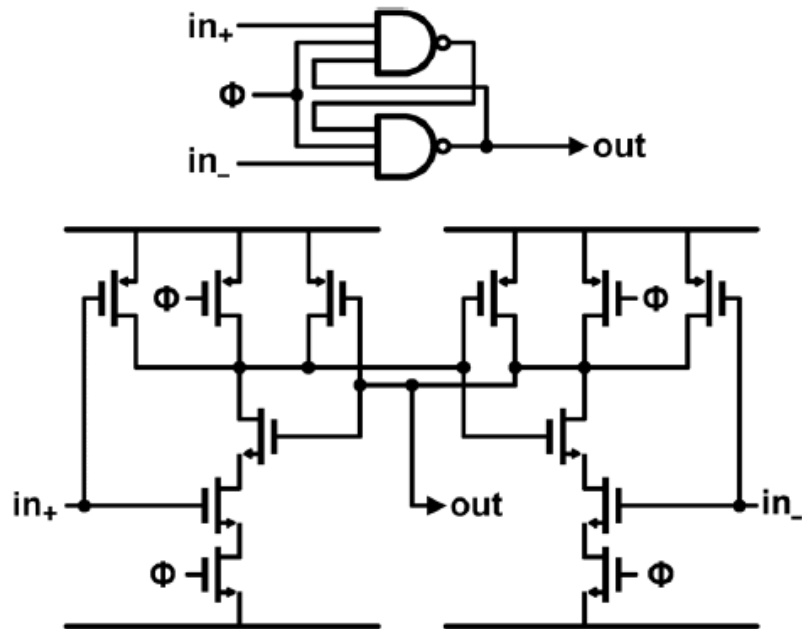


(c)



S. Weaver, IEEE TCAS-I, pp. 2825, 2010

# Stochastic Flash ADC (2)

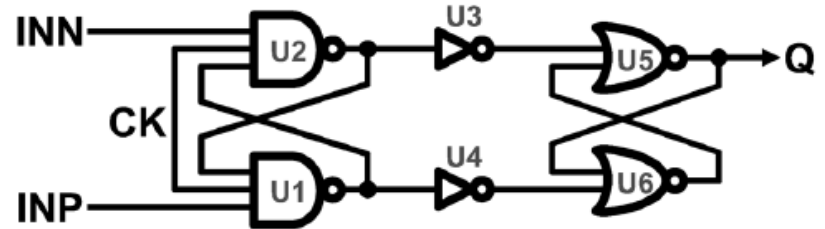


```

1  module comparator(INP, INN, CK, Q);
2  output Q;
3  input INP, INN, CK;
4
5  wire op, on, opn, onn, qn;
6  nand3x1 U1 ( .A(op), .B(INP), .C(CK), .Y(on) );
7  nand3x1 U2 ( .A(on), .B(INN), .C(CK), .Y(op) );
8  invx1 U3 ( .A(op), .Y(opn) );
9  invx1 U4 ( .A(on), .Y(onn) );
10 nor2x2 U5 ( .A(qn), .B(opn), .Y(Q) );
11 nor2x2 U6 ( .A(Q), .B(onn), .Y(qn) );
12
13 endmodule

```

(a)



(b)